Hyperspectral and LiDAR Classification via Frequency Domain-Based Network

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***Abstract*— Local–global feature learning method based on deep learning has significantly improved the collaborative classification of hyperspectral imagery (HSI) and light detection and ranging (LiDAR) data. However, HSI encompasses numerous bands with significant interband correlations. Addressing how to efficiently capture spatial, spectral, and elevation information from hyperspectral and LiDAR data while considering data redundancy and enhancing feature representation of land cover will contribute to enhancing classification effectiveness. Combining frequency feature learning methods with convolutional neural networks (CNNs), transformers, and other architectures to construct an end-to-end feature learning network framework is an effective method. Therefore, this article proposes a frequency domain-based network (FDNet) for the classification of HSI and LiDAR data using a frequency local feature learning framework and a self-attention mechanism based on fast Fourier transform (FFT). FDNet could effectively capture local efficient frequency features of spatial, spectral, and elevation information in HSI and LiDAR data in an adaptive feature learning style, and embedding convolutional offsets into the frequency domain-based transformer network not only enhances local features but also effectively captures global semantic characteristics of land covers while reducing computational complexity. We validated the efficacy of FDNet across three publicly available datasets and a particularly challenging self-constructed dataset, denoted as the Yancheng dataset. The source codes will be available at https://github.com/RSIP-NJUPT/FDNet.**

***Index Terms*— Classification, deep learning, frequency feature learning, hyperspectral imagery (HSI), light detection and ranging (LiDAR), local–global feature learning.**

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I. Introduction

HYPERSPECTRAL imagery (HSI) is characterized by rich spectral information and high spectral resolution. Consequently, it can accurately reflect the spectral characteristics of different land covers [1], [2], [3], [4], [5], [6]. Nevertheless, challenges such as spectral variability within the same land covers and variability under the same spectrum, degrade the classification accuracy of HSI data [7], [8], [9], [10], [11]. Light detection and ranging (LiDAR) systems can directly acquire high-precision, high-density 3-D spatial information of objects. The elevation information from LiDAR complements HSI data, leading to an enhancement in the accuracy of land cover classification [12], [13], [14], [15]. However, while considering the characteristics of the images, the collaborative classification of HSI and LiDAR data faces the following issues.

1) HSI exhibits a multitude of spectral bands with strong interband correlations. Thus, formulating an effective approach for local spatial–spectral feature learning style is crucial for HSI and LiDAR land classification.
2) The fusion of multisource features and the rapid capture of their global contextual information has a significant impact on enhancing the accuracy of land cover classification and its practical applications.

In previous work, numerous machine learning algorithms have been proposed, such as support vector machines (SVMs) [16] and random forest (RF) [17]. However, these methods mostly rely on the design of handcrafted features, resulting in limited classification accuracy. To leverage spatial information in HSI data, Zhang et al. [18] introduced a spatial-logical aggregation network with morphological transformation. To further enhance the fusion of multisource data,

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Gao et al. [19] developed a depthwise feature interaction network to capture self-correlation and cross correlation among multisource features. Hong et al. [20] proposed a shared and specific feature learning method to improve the integration of multimodal data. However, the aforementioned methods have limitations in capturing the complex relationship between spatial–spectral features from HSI and elevation features from LiDAR data in an effective local–global feature learning style.

In recent years, owing to the rapid advancement of deep learning, there have been a number of networks dedicated to learning local spatial and spectral features of HSI and LiDAR data [21], [22], [23], [24], [25], [26]. Hang et al. [27] proposed a coupled CNNs and several fusion strategies to improve joint classification accuracy. Ding et al. [28] designed a multiscale local spatial feature learning and aggregation module for joint classification of HSI and LiDAR data. Song et al. [29] incorporated transformer and convolutional neural networks (CNNs) to capture both the spectral and spatial characteristics of HSI and LiDAR data. For better describing the spatial, spectral, and elevation characteristics of land covers, Li et al. [30] utilized a three-branch neural network, named spatial–spectral saliency reinforcement network. The aforementioned methods predominantly focus on learning spatial and spectral features in the spatial domain. However, the feature learning method based on the frequency domain offers insights into image structures, such as edges, textures, and shapes. In addition, HSI encompasses numerous spectral bands, leading to data redundancy. Then, frequency domain feature learning not only facilitates effective feature extraction but also mitigates data redundancy, thereby enhancing feature representation capabilities. Consequently, frequency domain spatial–spectral feature learning stands as an effective approach for enhancing the performance of HSI and LiDAR land cover classification.

For the global feature learning style, the transformer model's robust capability in modeling long-range dependencies and learning contextual features has led to its widespread application in HSI and LiDAR for global feature learning of land cover characteristics [31], [32]. Zhao et al. [33] inserted a cross-token attention fusion encoder mechanism into the transformer for integrating the HSI and LiDAR features. Moreover, a novel vision transformer (ViT)-based baseline network, including HSI spectral transformer, HSI and LiDAR spatial transformer module, and collaboration transformer, was proposed to extract distinguishable features of HSI and LiDAR data [34]. Simultaneously, Xue et al. [35] designed a spectral sequence transformer module and a spatial hierarchical transformer module for acquiring global spectral relationships and hierarchical spatial features. Despite the superior classification performance achieved by the aforementioned transformer-based global feature learning methods, the scaled dot-product attention in these approaches typically requires complex matrix multiplication, resulting in quadratic space and time complexity. Recently, most methods have addressed this issue by employing the downsampling strategy to diminish the spatial resolution of features [36]. Nevertheless, this reduction in spatial resolution inevitably results in a loss of features. For land covers with complex scene information, the loss of these features may affect the accuracy of land cover classification. While significantly reducing computational complexity, the fast Fourier transform (FFT) can effectively capture features in the frequency domain, minimizing the loss of valuable features. Therefore, exploring a method that enables rapid computation of self-attention mechanisms based on the FFT algorithm presents an efficient approach to accurately characterize the properties of HSI and LiDAR data.

With the above motivation, we propose a frequency domain-based network, named FDNet, which can exploit the advantages of frequency domain feature learning to effectively capture both local and global features of spatial, spectral, and elevation information in HSI and LiDAR data. Considering the advantages of 2-D and 3-D wavelet feature learning in multiscale analysis and local feature acquisition, we devised a triple-branch network operating across the entire frequency spectrum. By integrating adaptive feature selection and learning mechanisms for both high-frequency and low-frequency components of wavelet transformations, we could achieve discernible effectiveness in capturing local features. Moreover, utilizing the advantages of the transformer-based global feature learning method, we leverage deformable convolutions (DConvs) to further comprehensively capture local features of fused characteristics. Simultaneously, we incorporate the FFT instead of matrix multiplication in the spatial domain to efficiently capture global fused features. In summary, our contributions are summarized as follows.

1) We propose a frequency local feature learning framework based on the discrete wavelet transform (DWT) method, including triple branch networks, i.e., spectral, spatial, and elevation learning and enhancement modules. This framework conducts spatial feature learning for low-frequency components and adaptive feature learning for high-frequency components. This approach effectively enhances the discriminability of local features.
2) We develop a self-attention mechanism based on the FFT, which facilitates simultaneous global feature learning in the frequency domain while effectively reducing the computational complexity of matrix multiplication. In addition, embedding convolutional offsets enhances the discernibility of fused local features.
3) We analyze the effectiveness of FDNets in joint HSI and LiDAR classification. It is noteworthy that we construct a more challenging dataset, namely, the Yancheng dataset, and validate the effectiveness of the proposed FDNet on this dataset. Furthermore, we have made the code and model zoo publicly available.

II. Related Work

A. Local–Global Feature Learning for HSI and LiDAR Classification

Local and global features play significant roles in HSI and LiDAR for land cover classification. Global features exhibit superior performance in handling relatively uniform land cover categories, whereas local features excel in dealing with land cover types characterized by intricate textures and diverse structures. In recent years, a number of relevant algorithms have been successively proposed. For local

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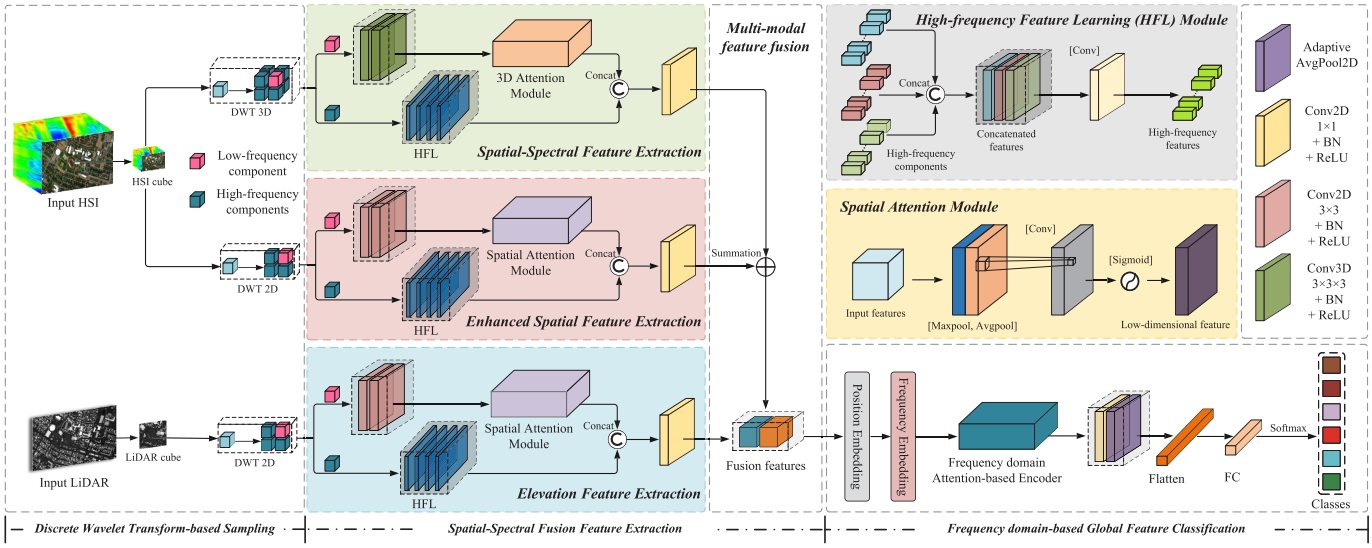


Fig. 1. Overall framework of FDNet for HSI and LiDAR classification, including DWTS, S2FFE, and FGFC. S2FFE consists of three main parts: spatial–spectral feature extraction, enhanced spatial feature extraction, and elevation feature extraction, in which HFL represents the high-frequency feature learning module.

features, Peng et al. [37] started from a multiview strategy and employed Gabor filters and ResNet-like CNNs for capturing spatial–spectral information of the fusion feature. Similarly, Feng et al. [9] adopted a multiview feature learning approach, embedding it into a graph CNN. This integration encompassed spectral, spatial, and elevation features, aiming to acquire superior classification performance. Zhang et al. [38] paid much attention to information fusion style and CNNs, and proposed an interleaving perception CNN for HSI and LiDAR classification. Other related works, such as multiscale deep learning network with self-calibrated convolution network [39], a novel three-branch CNNs [40], patch-to-patch CNN [41], dual-branch dynamic modulation network [42], and morphological convolution [43]. The majority of the aforementioned methods belong to the spatial domain local feature learning framework, failing to adequately consider the advantages of frequency domain local feature learning.

In the global feature learning of HSI and LiDAR data, the application of contextual feature learning, long–short-term memory (LSTM) network, and transformer is notably extensive, e.g., [28], [33], [34], [35], [44], [45], [46], [47]. Enhancements in most algorithms are concentrated on the integration of cross-self-attention mechanisms for multisource fusion features, as well as the combination of convolutional and LSTM networks, categorizing them under spatial domain feature learning approaches. Nevertheless, these enhancements have not paid much attention to the computational complexity of the self-attention mechanism and fail to recognize the critical importance of global feature learning within the frequency domain.

B. Transformer

The transformer [48] has the capability to capture global contextual information, resulting in notable advancements across a wide range of advanced vision tasks. To reduce the computational cost of the transformer, Fan et al. [49] inserted a multihead pooling attention mechanism into the transformer to reduce the spatial dimension and facilitate the efficient representation of complex visual data. Zamir et al. [50] reduced spatial dimensions hierarchically within the encoder to mitigate subsequent computational complexity in the self-attention mechanism. Moreover, there exist a number of works addressing adaptive and sparse transformers, such as [51], [52], [53], [54], [55], and [56]. The aforementioned approach leverages sparse self-attention or adaptive attention mechanisms, substituting the conventional dense attention of the transformer, thereby reducing computational complexity. In particular, Kong et al. [57] develop a self-attention solver based on frequency domain optimization (FFT) for enhanced efficiency. However, when it comes to integrating HSI and LiDAR features, this method utilizes point-wise convolution and depth-wise convolution to learn local features. Nonetheless, it may struggle to effectively adapt to features of different sizes and shapes in complex land cover structures. Therefore, achieving an effective representation of locally fused features before frequency-domain global feature learning could significantly enhance the classification performance of HSI and LiDAR data.

III. Methodology

The proposed FDNet is designed for pixel-level land cover classification by capturing deep fusion features of HSI and LiDAR data in the frequency domain. Fig. 1 illustrates the general structure of the proposed method, which consists of three main parts: DWT-based sampling (DWTS), spatial–spectral fusion feature extraction (S2FFE), and frequency domain-based global feature classification (FGFC). DWTS utilizes DWT to transform the input HSI and LiDAR data into different frequency intervals and simultaneously down-samples them to acquire the low and high-frequency components of the original data. S2FFE, based on 3-D CNNs and 2-D CNNs, learns deep features from the preserved

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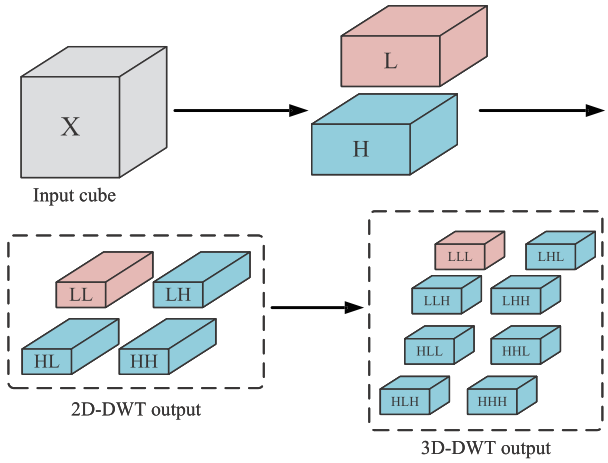


Fig. 2. Illustration of DWT.

low-frequency components containing the main information and learns detailed features from the high-frequency components to fully integrate the spatial–spectral features in HSI data and the elevation information in LiDAR data. FGFC, based on the designed frequency domain attention-based encoder (FAE), models long-range dependency relationships to improve the model's classification accuracy via global features.

For the original HSI data $\mathbf{X}_H \in \mathbb{R}^{M\times N\times d_h}$ and LiDAR data $\mathbf{X}_L \in \mathbb{R}^{M\times N}$, where M and N refer to the original height and width of HSI and LiDAR data, respectively. d_h denotes the original number of spectral bands of HSI. After padding the edge pixels, patches are extracted for each pixel, resulting in HSI cubes $\mathbf{X}_H^P \in \mathbb{R}^{p\times p\times d_h}$ and LiDAR cubes $\mathbf{X}_L^P \in \mathbb{R}^{p\times p}$, where $p \times p$ denotes the spatial size. Then, $\mathbf{X}_H^P$ and $\mathbf{X}_L^P$ are employed as the input data.

A. DWT-Based Sampling

In joint land cover classification tasks incorporating HSI and LiDAR data, the majority of previous approaches directly utilized the application of CNNs to capture the semantic features of $\mathbf{X}_H^P$ and $\mathbf{X}_L^P$. Nevertheless, CNNs are often susceptible to noise interference, and lacking data filtering before downsampling through operations like max-pooling or average-pooling in numerous CNNs-based methods may result in noise from high-frequency components being downsampled into the output features, affecting the model's classification accuracy. Considering that the DWT can preserve high-frequency details during downsampling, it provides a richer feature representation, enabling models to better capture subtle variations and features in the data. In addition, the capability of DWT to decompose HSI and LiDAR data into different frequency wavelet components helps the model better understand the frequency characteristics of the data, thereby enhancing the model's adaptability to complex scenes. Hence, we improve a DWT layer suitable for various wavelets to replace traditional downsampling operations, decomposing the data into low-frequency and high-frequency components during the downsampling stage and inserting them into CNNs to learn robust deep features.

For the input HSI cubes $\mathbf{X}_H^P$, 2D-DWT and 3D-DWT are utilized to perform downsampling operations. As shown in Fig. 2, the 2D-DWT produces one low-frequency component $\mathbf{X}_H^{ll}$ containing the main structural information and three high-frequency components $\mathbf{X}_H^{lh}$, $\mathbf{X}_H^{hl}$, and $\mathbf{X}_H^{hh}$ including noise and image details. 3D-DWT produces one low-frequency component $\mathbf{X}_H^{lll}$ and seven corresponding high-frequency components [58]. Given that with a slight modification based on 2D-DWT, it can be extended to 3D-DWT. Then, the 2D-DWT components of each frequency can be calculated via

$$\begin{aligned} \mathbf{X}_H^{ll} &= \mathbf{L}\mathbf{X}_H^P\mathbf{L}^T, \quad \mathbf{X}_H^{lh} = \mathbf{H}\mathbf{X}_H^P\mathbf{L}^T \\ \mathbf{X}_H^{hl} &= \mathbf{L}\mathbf{X}_H^P\mathbf{H}^T, \quad \mathbf{X}_H^{hh} = \mathbf{H}\mathbf{X}_H^P\mathbf{H}^T \end{aligned} \tag{1}$$

where $\mathbf{L} = \{l_k\}_{k\in\mathbb{Z}}$ and $\mathbf{H} = \{h_k\}_{k\in\mathbb{Z}}$ represent the low-pass and high-pass filter, respectively. Similarly, for the input LiDAR cubes $\mathbf{X}_L^P$, 2D-DWT is performed to downsample and acquire one low-frequency component $\mathbf{X}_L^{ll}$ and three high-frequency components $\mathbf{X}_L^{lh}$, $\mathbf{X}_L^{hl}$, and $\mathbf{X}_L^{hh}$.

B. Spatial–Spectral Fusion Feature Extraction

The proposed S2FFE aims to fully utilize the rich spatial–spectral information in HSI data and the elevation information in LiDAR data. As shown in Fig. 1, the module mainly consists of spatial–spectral feature extraction, enhanced spatial feature extraction, and elevation feature extraction, and these three parts have the same structure: independent processing the low-frequency and high-frequency components output by DWTS with targeted branches, and then merging the high and low-frequency features. Specifically, for the low-frequency components including the structural features, deep feature extraction is directly applied; for the high-frequency components, a simple and efficient high-frequency feature learning (HFL) module is designed to capture the detailed features within them.

Benefiting from the unique properties of 3-D CNNs that can capture both spectral and spatial information, along with the strong local feature learning capability of 2-D CNNs, we utilize them in the feature extraction branch of low-frequency components of HSI and LiDAR data. The aim is to enhance the model's local spatial–spectral and elevation learning ability for multimodal data. For the HSI low-frequency components $\mathbf{X}_H^{lll} \in \mathbb{R}^{(p/2)\times(p/2)\times(d_h/2)}$ and $\mathbf{X}_H^{ll} \in \mathbb{R}^{(p/2)\times(p/2)\times d_h}$ acquired by DWTS, the encoded features through two layers of convolution can be obtained by

$$\begin{aligned} \mathbf{f}_H^{lll} &= \mathrm{Conv}_{3\times3\times3}^{16}\left(\mathrm{Conv}_{3\times3\times3}^{8}\left(\mathbf{X}_H^{lll}\right)\right) \\ \mathbf{f}_H^{ll} &= \mathrm{Conv}_{3\times3}^{64}\left(\mathrm{Conv}_{3\times3}^{32}\left(\mathbf{X}_H^{ll}\right)\right) \end{aligned} \tag{2}$$

where $\mathrm{Conv}_{3\times3\times3}^{16}$ represents a convolutional network with a kernel size of $3 \times 3 \times 3$, 16 kernels, and cascaded with a batch normalization layer and ReLU activation operation. By applying similar operations to the LiDAR low-frequency component $\mathbf{X}_L^{ll} \in \mathbb{R}^{(p/2)\times(p/2)}$, one can acquire $\mathbf{f}_L^{ll}$.

It is worth noting that a spatial attention (SA) module is introduced to selectively learn important features for classification tasks while reducing the focus on irrelevant information. The aim is to adaptively learn important features through the module to enhance the effectiveness of low-frequency features.

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For the input convolutional encoded features, the attention map of the same size as the input feature is generated based on the spatial relationships of the features to adaptively allocate the attention of the features. Specifically, we apply max-pooling and average-pooling operations along the spectral channel dimension of input features, concatenate to obtain the SA map, and then apply element-wise multiplication between the input feature map and the attention map. For SA of HSI data, spatial attention module (AM) can be obtained via

$$\mathbf{M}_H^{ll} = \sigma\left(f^{a\times a}\left(\left[\text{MaxPool}\left(\mathbf{f}_H^{ll}\right); \text{AvgPool}\left(\mathbf{f}_H^{ll}\right)\right]\right)\right) \quad (3)$$

$$\mathbf{F}_H^{ll} = \mathbf{M}_H^{ll} \odot \mathbf{f}_H^{ll} \quad (4)$$

where σ represents the sigmoid function, $\odot$ denotes element-wise multiplication, and $f^{a\times a}$ represents a convolution operation with a kernel size of $a \times a$. Through similar operations, low-frequency elevation features of LiDAR data $\mathbf{M}_L^{ll}$ and low-frequency 3-D spatial–spectral features of HSI data $\mathbf{M}_H^{lll}$ can be acquired. Notably, for 3-D SA of HSI data, the input feature map of 3-D AM has one more dimension than the input feature map of the SA module, and it replaces the 2-D CNNs in SA with 3-D CNNs. Therefore, 3-D AM can operate simultaneously in the spectral and spatial dimensions. For HSI data rich in spectral information, 3-D AM comprehensively captures interband relationships and pixel spatial correlations. By dynamically adjusting attention weights based on spectral and spatial features at each pixel, important features are emphasized while redundant features are reduced.

Compared to global operations, local operations excel at preserving subtle structures and texture information, making them more suitable for handling high-frequency detailed information in images. Therefore, while considering model complexity, use convolution operations with a kernel size of 1 to learn detailed features in high-frequency components. Specifically, concatenate all high-frequency components output by DWTS along the channel dimension, and after convolution processing, detailed high-frequency features can be obtained via

$$\mathbf{F}_H^{hh} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{X}_H^{lh}, \mathbf{X}_H^{hl}, \mathbf{X}_H^{hh}\right)\right). \quad (5)$$

After obtaining the high-frequency features $\mathbf{F}_L^{hh}$ of LiDAR data learned by similar operations and the HSI spectral features $\mathbf{F}_H^{hhh}$, concatenate the high-frequency features with the corresponding low-frequency features along the channel dimension, map them to a unified channel dimension through convolutional layers, and obtain spatial features $\mathbf{F}_H^{2\text{-d}}$, spectral features $\mathbf{F}_H^{3\text{-d}}$ based on HSI data, and elevation features $\mathbf{F}_L^{2\text{-d}}$ based on LiDAR data. Then, the multimodal fusion features $\mathbf{F}_{\text{cnn}}$ extracted based on CNNs can be represented as

$$\mathbf{F}_{\text{cnn}} = \omega \cdot \left(\mathbf{F}_H^{2\text{-d}} \oplus \mathbf{F}_H^{3\text{-d}}\right) \oplus (1-\omega) \cdot \mathbf{F}_L^{2\text{-d}} \quad (6)$$

where $\oplus$ represents the element-wise summation operation and ω is the weight coefficient, which can be manually adjusted.

C. Frequency Domain-Based Global Feature Classification

Although 3-D CNNs provide a richer spectral feature representation, CNNs may still be constrained by its hierarchical structure and lack global contextual features. In contrast, a ViT, based on multihead self-attention (MHSA), is capable of capturing interband correlations on a global scale and efficiently handling multimodal data, integrating them into a unified feature representation. Furthermore, since the calculation of scaled dot-product attention can be implemented through convolution operations when tokens are rearranged, and spatial domain convolution is equivalent to element-wise multiplication in the frequency domain. Therefore, as shown in Fig. 3, we develop an efficient FAE to integrate multimodal fusion features from $\mathbf{F}_{\text{cnn}}$ and perform global feature extraction. Specifically, FAE estimates the correlation between one token in the query and all tokens in the key using convolution operation, and applies a replacement technique where spatial domain-based convolution operation equals point-wise multiplication in the frequency domain, effectively reducing the computation of the scaled dot-product attention in the transformer: from the original quadratic space and time complexity of the number of tokens to $O(N)$ and $O(N\log N)$, respectively, where N represents the number of pixels.

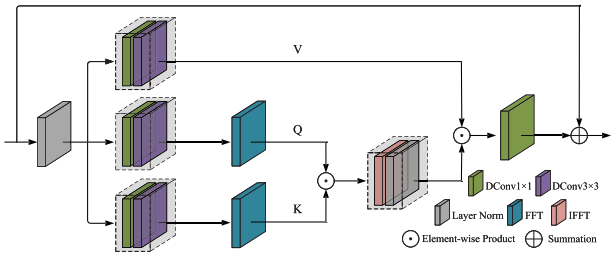


Fig. 3. Illustration of FAE. The DConv is adopted to replace the linear layer in the original MHSA mechanism, and then the feature maps are transformed to the frequency domain using FFT, achieving $\mathbf{QK}^{\mathsf{T}}$ based on element-wise multiplication. While obtaining global contextual features, it effectively avoids complex matrix multiplication, reducing computational complexity. See the texts for further details.

In the spatial domain, a traditional attention map is based on the query $\mathbf{Q}$, key $\mathbf{K}$, and value $\mathbf{V}$ obtained by the input sequence passing through three linear mappings, as shown

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{QK}^{\mathsf{T}}}{\sqrt{d_k}}\right)\mathbf{V}. \quad (7)$$

Considering complex matrix multiplication of $\mathbf{QK}^T$, the designed FAE employs convolution operation instead of linear transformations, acquiring the corresponding $\mathbf{Q}_f$, $\mathbf{K}_f$, and $\mathbf{V}_f$. Then, FFT is applied to $\mathbf{Q}_f$ and $\mathbf{K}_f$, followed by obtaining the global aggregated features via

$$\mathbf{V}_{\text{attn}} = \mathcal{L}(\mathcal{F}^{-1}(\mathcal{F}(\mathbf{Q}_f)\mathcal{F}(\mathbf{K}_f)^{\mathsf{T}}))\mathbf{V}_f \quad (8)$$

$$\mathbf{F}_{\text{global}} = \mathbf{F}_f + \text{Conv}_{1\times1}(\mathbf{V}_{\text{attn}}) \quad (9)$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ represent FFT and fast Fourier inverse transform (IFFT), respectively [57]. $\mathcal{L}(\cdot)$ denotes layer normalization. For $\mathbf{F}_f$, after the input multimodal fusion feature $\mathbf{F}_{\text{cnn}}$ feeds into the common position embedding, each point consists of the relative positional information. Since FAE operates in the frequency domain, frequency embedding [FE(·)] is proposed to embed frequency information, with the calculation as follows:

$$\text{FE}_{(j,k)}(\cdot) = \sin(j/10\,000^{k/c}) \quad (10)$$

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where j is the jth pixel of feature map $\mathbf{F}_{\text{cnn}}$, and k represents the kth channel in all channels c of input features. Position embedding operation is recorded as $\mathcal{P}(\cdot)$, then the input feature of FAE can be obtained by

$$\mathbf{F}_f = \text{FE}(\mathcal{P}(\mathbf{F}_{\text{cnn}})). \tag{11}$$

Due to the fixed shape and sampling positions of traditional convolutional kernels, and although stacking multiple kernels to form a convolutional chain can broaden the receptive field, the attention of each kernel is limited to a finite region, thus this method is restrictive. Thus, we replace traditional convolution with DConv with the same kernel parameter settings for the convolutional operations in FAE. The aim is to utilize the ability of DConv to dynamically adjust sampling positions by adapting offsets according to input features, thus expanding the receptive field [59], better perceiving the spatial structure of input features, and capturing global information from input features by FAE.

Specifically, the offset is obtained by performing an additional convolution operation on the input features. Since each pixel has two offsets, one in the horizontal direction and the other in the vertical direction. Then, for the input features $\mathbf{X}$, the offset can be calculated by

$$\Delta\mathbf{X} = \text{Conv}_{k\times k}^{2k^2}(\mathbf{X}, \theta) \tag{12}$$

where k represents the size of the convolution kernel used for offset calculation, and θ denotes other convolutional kernel parameters involved in offset computation. After adjusting the sampling position based on the offset, the output can be obtained by

$$\mathbf{Y}(p_0) = \sum_{p_n\in\mathcal{R}} \mathbf{W}(p_n)\cdot\mathbf{X}(p_0 + p_n + \Delta\mathbf{X}(p_n)) \tag{13}$$

where p_0 represents any position on output features and p_n denotes the enumeration of positions listed in the sampling grid $\mathcal{R}$.

Finally, followed by the global frequency-domain feature $\mathbf{F}_{\text{global}}$ extracted by FAE, a CNN-based classifier is designed for classification. Specifically, $\mathbf{F}_{\text{global}}$ is sequentially processed through 2-D CNN, adaptive global average pooling, fully connected layer, and softmax function to obtain the final probability vector $\mathbf{P}_f \in \mathbb{R}^{1\times\text{cls}}$, where the label with the highest probability represents the category of land covers.

IV. Experiments

In the experiments, four HSI and LiDAR datasets, including a more challenging Yancheng dataset that we built, are utilized to evaluate the effectiveness of FDNet and compare it with other advanced networks.

A. Data Description

1) Yancheng Dataset: The Yancheng dataset includes HSI and a LiDAR-based digital surface model (DSM). It was captured on October 15, 2022, over Yancheng City, China, at a flight altitude of 600 m. The HSI data was collected using a visible light imaging spectrometer from the multiparameter airborne remote sensing system (MARSS), containing

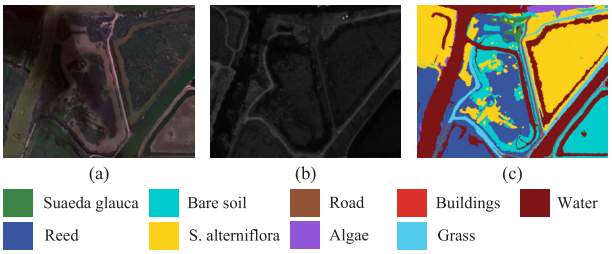


Fig. 4. Illustration of Yancheng dataset. (a) Pseudo-color composite image based on bands 40, 60, and 120 for HSI. (b) LiDAR-based DSM. (c) Ground-truth.

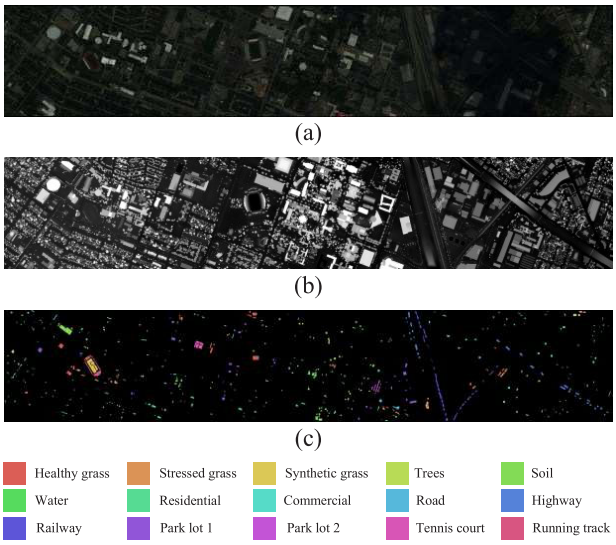


Fig. 5. Illustration of Houston2013 dataset. (a) Pseudo-color composite image based on bands 64, 43, and 20 for HSI. (b) LiDAR-based DSM. (c) Ground-truth.

249 spectral bands with a wavelength range from 400 to 1000 nm. To reduce redundant information and computational costs, principal component analysis (PCA) was used to reduce the number of spectral channels in the HSI data to 30. The LiDAR data was obtained by the RIEGL VUX-240 laser scanner, with a single spectral band and an average point density of 51.44 pts/m^2. The dimensions of the HSI and LiDAR data are 400 × 500 pixels, with a spatial resolution of 1 m. The dataset comprises nine different land cover categories, totaling 200 000 real samples. Table I lists the training and testing samples and Fig. 4 shows a pseudo-color composite image of the HSI data, a grayscale image of the LiDAR data, and the ground-truth.

2) Houston2013 Dataset: The Houston2013 dataset comprises an HSI and a LiDAR-based DSM, acquired by the National Center for Airborne Laser Mapping (NCALM) in June 2012 utilizing the ITRES CASI-1500 imaging sensor over the University of Houston campus. It was provided by the IEEE GRSS Data Fusion Competition. The HSI encompasses 144 spectral bands spanning a wavelength range from 0.38 to 1.05 μm, while LiDAR data are represented by a single band. Both the HSI and LiDAR datasets share dimensions of 349 × 1905 pixels, with a spatial resolution of 2.5 m. The dataset contains 15 distinct categories, amounting to a total

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TABLE I
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR YANCHENG DATASET (BEST RESULTS ARE IN BOLD)

No.	Class(Train/Test)	CoupledCNN	EndNet	MDL-Cross	MS2CANet	S2ENet	MFT	NNCNet	HCTNet	ExViT	FDNet
1	Suaeda glauca (20/2520)	97.18	87.30	91.94	**99.13**	97.46	94.04	96.47	94.56	96.27	98.25
2	Bare soil (20/30275)	72.09	56.31	67.68	**74.47**	73.05	55.56	60.89	24.66	65.66	71.65
3	Road (20/2956)	82.81	68.54	**87.75**	80.72	78.76	87.01	82.58	77.33	78.59	81.22
4	Buildings (20/87)	**100**	79.31	83.91	**100**	98.85	**100**	97.70	**100**	96.55	**100**
5	Water (20/49297)	85.72	77.06	71.52	84.41	85.16	**88.51**	70.42	77.85	78.78	88.18
6	Reed (20/42251)	53.04	48.04	55.56	64.92	68.78	**72.45**	67.94	22.43	62.35	68.97
7	S. alterniflora (20/58880)	56.94	56.35	79.66	70.48	66.62	58.42	72.26	**82.81**	78.26	73.80
8	Algae (20/3282)	88.18	76.87	93.24	93.57	85.59	86.99	93.42	89.52	89.58	**94.94**
9	Grass (20/10272)	68.33	52.44	66.22	67.70	54.49	68.53	67.96	69.83	67.28	**71.38**
	OA(%)	67.52	60.41	70.55	74.11	72.90	70.26	69.76	59.53	72.98	**76.65**
	AA(%)	78.26	66.91	77.50	81.71	78.75	79.06	78.85	71.00	79.26	**83.15**
	Kappa(%)	60.74	51.27	63.13	68.01	66.38	63.33	62.36	49.20	66.12	**71.03**

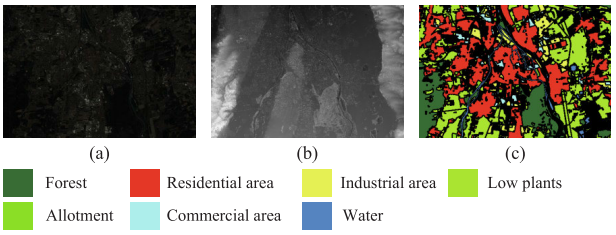


Fig. 6. Illustration of Augsburg dataset. (a) Pseudo-color composite image based on bands 20, 15, and 10 for HSI. (b) LiDAR-based DSM. (c) Ground-truth.

of 15 029 real samples [28]. Table II shows the training and testing samples and Fig. 5(a)–(c) illustrates a pseudo-color composite image of the HSI, a grayscale image of the LiDAR, and the ground-truth, respectively.

3) Augsburg Dataset: The Augsburg dataset was captured over Augsburg, Germany. HSI data were gathered using the DAS-EOC HySpex sensor, while LiDAR-based DSM data were acquired through the DLR-3K system. To facilitate multimodal fusion, both images were down-sampled to a uniform resolution of 30 m. This dataset contains HSI data with 180 bands ranging from 0.4 to 2.5 μm and DSM data in a single band [33]. With dimensions of 332 $\times$ 485 pixels, the dataset represents seven distinct land cover categories. Table III shows the number of training and testing samples we used, and illustrated in Fig. 6(a)–(c) are a pseudo-color composite image of the HSI, a grayscale image of the LiDAR, and the ground-truth, respectively.

4) MUUFL Dataset: The MUUFL dataset was obtained in November 2010 at the University of Southern Mississippi Gulf Park campus in Long Beach, MS, USA. HSI was collected using the ITRES Research Limited (ITRES) Compact Airborne Spectral Imager (CASI-1500) sensor, initially comprising 72 bands [12]. Due to excessive noise, the first and last eight spectral bands were eliminated, resulting in a total of 64 available spectral channels covering wavelengths from 0.38 to 1.05 μm. LiDAR data was captured by an ALTM sensor, consisting of two rasters at a wavelength of

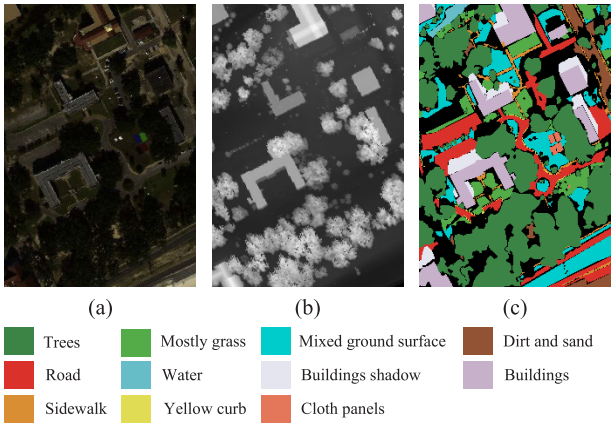


Fig. 7. Illustration of MUUFL dataset. (a) Pseudo-color composite image based on bands 20, 15, and 5 for HSI. (b) LiDAR-based DSM. (c) Ground-truth.

1.06 μm. The dataset comprises 53 687 ground-truth pixels, representing 11 distinct land cover classes. Table IV shows the training and testing samples. In Fig. 7(a)–(c), a pseudo-color composite image representing the HSI data, a grayscale depiction of LiDAR data, and a ground-truth are presented, respectively.

B. Experimental Setting

The proposed FDNet is implemented in the PyTorch framework. The experiments are performed on the Ubuntu 22.04 platform equipped with an I9-13900K CPU, an NVIDIA RTX 4090Ti GPU, and RAM: 32 GB. An AdamW optimizer is adopted to optimize the network. In the training stage, batch size, and the number of training epochs are set to 64, and 500, respectively. The initial learning rate is set to $1e^{-}3$ and adjusted through a cosine annealing scheme. In terms of model parameter configuration, the depths of FAE and patch size are set to 1 and 8, respectively. And initial values of the feature fusion weight coefficient ω are set to 0.7. Daubechies wavelet with order 2 (db2) is utilized to perform DWT, while standard cross-entropy is utilized as the loss function.

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TABLE II
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR HOUSTON2013 DATASET (BEST RESULTS ARE IN BOLD)

No.	Class(Train/Test)	CoupledCNN	EndNet	MDL-Cross	MS2CANet	S2ENet	MFT	NNCNet	HCTNet	ExViT	FDNet
1	Healthy grass (20/1231)	85.70	85.05	93.18	88.38	83.51	93.50	86.92	95.69	96.51	**97.32**
2	Stressed grass (20/1234)	94.17	96.27	97.33	97.00	95.54	94.65	95.70	98.46	95.14	**98.78**
3	Synthetic grass (20/677)	91.14	99.70	99.56	98.08	98.82	98.97	**100**	99.85	93.94	97.49
4	Trees (20/1224)	69.53	94.36	**99.84**	96.65	99.26	98.37	92.97	97.96	**99.84**	**99.84**
5	Soil (20/1222)	99.92	99.92	98.36	**100**	**100**	97.95	**100**	**100**	98.94	**100**
6	Water (20/305)	97.05	97.38	98.36	99.34	**100**	**100**	98.03	98.03	99.67	96.39
7	Residential (20/1248)	92.07	89.90	90.63	84.38	97.84	81.65	94.15	**97.92**	91.67	96.23
8	Commercial (20/1224)	64.62	67.48	86.76	66.01	91.91	90.44	80.56	77.86	**94.61**	89.54
9	Road (20/1232)	93.34	86.04	**94.16**	68.34	91.56	84.74	68.18	83.52	88.88	84.50
10	Highway (20/1207)	72.99	85.09	81.03	97.76	92.29	94.28	96.85	97.18	87.90	**98.18**
11	Railway (20/1215)	33.50	87.98	92.10	75.80	**97.45**	94.81	93.09	93.74	95.97	96.30
12	Park lot 1 (20/1213)	77.33	79.97	79.80	92.66	92.00	89.28	82.28	92.66	90.68	**95.47**
13	Park lot 2 (20/449)	91.09	46.77	87.97	94.21	97.10	96.88	91.98	**99.55**	97.10	91.98
14	Tennis court (20/408)	99.75	98.28	98.53	**100**	**100**	99.75	**100**	**100**	**100**	**100**
15	Running track (20/640)	**100**	98.59	99.38	**100**	**100**	99.06	99.84	**100**	**100**	**100**
	OA(%)	81.30	87.58	92.31	88.64	94.98	93.11	90.61	94.53	94.65	**95.91**
	AA(%)	84.15	87.52	93.13	90.58	95.82	94.29	92.04	95.50	95.39	**96.13**
	Kappa(%)	79.77	86.56	91.69	87.73	94.57	92.55	89.85	94.09	94.22	**95.58**

TABLE III
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR AUGSBURG DATASET (BEST RESULTS ARE IN BOLD)

No.	Class(Train/Test)	CoupledCNN	EndNet	MDL-Cross	MS2CANet	S2ENet	MFT	NNCNet	HCTNet	ExViT	FDNet
1	Forest (20/13487)	97.42	92.91	**99.07**	96.63	98.43	94.88	96.26	95.37	94.62	98.12
2	Residential area (20/30309)	68.75	32.68	52.43	71.36	65.58	63.91	32.92	65.87	75.89	**78.26**
3	Industrial area (20/3831)	**93.08**	48.26	56.62	66.80	53.09	30.91	84.81	39.08	54.03	52.60
4	Low plants (20/26837)	75.30	66.92	84.00	54.43	80.04	63.64	64.44	**87.03**	81.07	85.39
5	Allotment (20/555)	95.86	68.47	83.96	93.69	96.22	84.86	**97.66**	94.95	81.98	82.34
6	Commercial area (20/1625)	2.15	30.22	52.37	43.26	51.45	**75.69**	10.03	63.38	59.51	58.71
7	Water (20/1510)	62.58	73.18	57.22	**72.91**	69.87	66.75	47.09	62.05	66.75	67.75
	OA(%)	75.83	56.58	71.84	69.29	75.61	67.99	57.47	76.99	79.35	**82.30**
	AA(%)	70.74	58.95	69.38	71.30	73.52	68.66	61.89	72.53	73.41	**74.74**
	Kappa(%)	68.15	45.91	63.61	59.87	67.76	58.24	48.25	69.58	71.85	**76.00**

Moreover, with the purpose of objectively evaluating the classification performance of the proposed FDNet and other compared networks, three widely used evaluation metrics are adopted: overall accuracy (OA), average accuracy (AA), and Kappa coefficient.

C. Performance Comparison

Several representative methods used for HSI and LiDAR joint classification are adopted to validate the effectiveness of FDNet, including CoupledCNN [27], EndNet [60], MDL-Cross [61], multiscale spatial–spectral cross-modal attention network (MS2CANet) [22], spatial–spectral cross-modal enhancement network (S2ENet) [62], multimodal fusion transformer (MFT) [63], NNCNet [12], HCTNet [33], extended ViT (ExViT) [64]. Among these methods, CoupledCNN, EndNet, MDL-Cross, MS2CANet, and S2ENet are all based on deep CNNs architectures, while MFT, NNCNet, HCTNet, and ExViT utilize CNNs-transformer architectures. The parameters of these algorithms are set based on their respective references. In addition, the same training and testing samples are randomly selected for fair comparison.

Several comparative methods are evaluated via visual comparisons (depicted in Figs. 8–11) and quantitative metrics, such as OA, AA, Kappa coefficient, and per-class accuracy. Tables I–IV clearly present the objective classification results of FDNet and each comparative method across the Yancheng, Houston2013, Augsburg, and MUUFL datasets, with the best results in each row highlighted in bold.

1) Quantitative Analysis: Tables I–IV present the quantitative classification results of various methods across four challenge HSI-LiDAR datasets. One can observe that our proposed FDNet demonstrates superior performance, obtaining the highest classification scores across all datasets. Specifically, our model outperforms the suboptimal model by 2.54%, 0.93%, 2.95%, and 1.87% in OA on the Yancheng, Houston2013,

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TABLE IV
COMPARISON OF CLASSIFICATION PERFORMANCES OBTAINED BY DIFFERENT METHODS FOR MUUFL DATASET (BEST RESULTS ARE IN BOLD)

No.	Class(Train/Test)	CoupledCNN	EndNet	MDL-Cross	MS2CANet	S2ENet	MFT	NNCNet	HCTNet	ExViT	FDNet
1	Trees (60/23186)	76.97	86.26	88.14	88.81	90.58	87.00	86.96	85.06	86.69	**90.60**
2	Mostly grass (60/4210)	45.46	78.98	62.66	**87.05**	81.38	83.68	60.88	76.86	82.52	83.28
3	Mixed ground surface (60/6822)	**88.68**	57.68	76.53	62.68	71.55	60.60	70.20	72.19	70.98	75.48
4	Dirt and sand (60/1766)	93.09	79.33	88.11	92.58	92.98	80.63	**97.79**	93.37	91.73	89.01
5	Road (60/6627)	67.60	83.99	86.00	79.92	80.34	82.57	72.58	75.80	83.99	**87.94**
6	Water (60/406)	99.75	98.03	**100**	**100**	**100**	**100**	**100**	**100**	99.75	**100**
7	Buildings shadow (60/2173)	75.79	87.53	94.62	90.38	**95.08**	92.27	75.38	93.83	93.56	92.50
8	Buildings (60/6180)	60.78	89.63	93.88	85.34	92.39	90.87	94.32	93.38	93.20	**95.87**
9	Sidewalk (60/1325)	65.58	62.11	62.42	65.13	72.38	74.19	58.19	67.47	**77.58**	75.62
10	Yellow curb (60/123)	91.87	86.99	86.99	**96.75**	93.50	91.06	68.29	87.80	87.80	93.50
11	Cloth panels (60/209)	98.56	96.65	98.09	98.09	**99.52**	**99.52**	98.09	**99.52**	99.04	98.56
	OA(%)	73.42	81.46	84.77	83.53	86.25	83.08	81.06	82.94	85.13	**88.12**
	AA(%)	78.56	82.47	85.22	86.07	88.15	85.67	80.24	85.94	87.90	**89.31**
	Kappa(%)	66.36	76.13	80.31	78.76	82.21	78.30	75.64	78.16	80.91	**84.58**

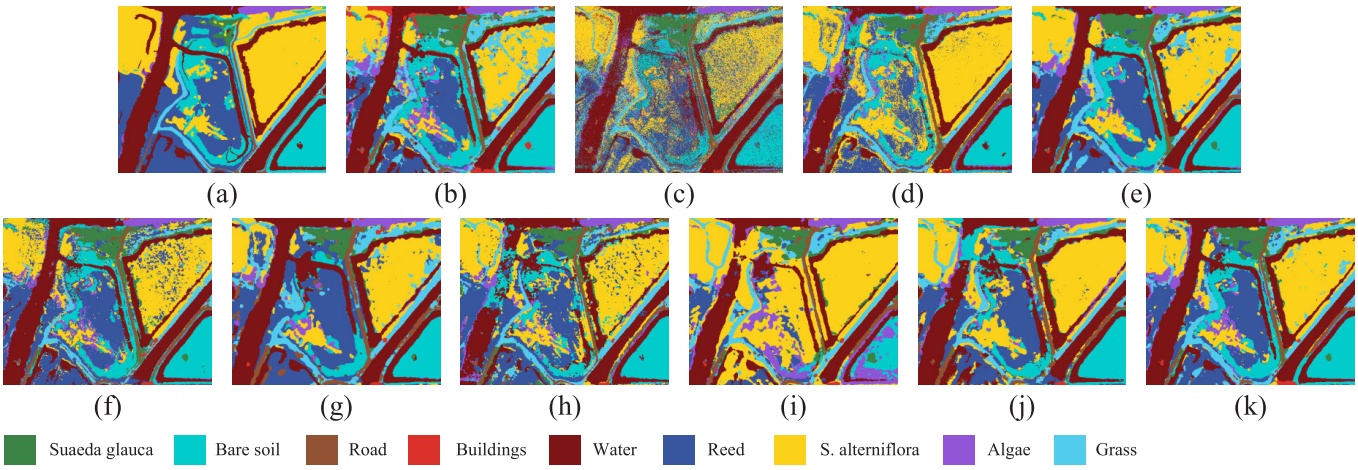


Fig. 8. Classification maps by different methods on Yancheng dataset. (a) Ground-truth. (b) CoupledCNN (67.52%). (c) EndNet (60.41%). (d) MDL-Cross (70.55%). (e) MS2CANet (74.11%). (f) S2ENet (72.90%). (g) MFT (70.26%). (h) NNCNet (69.76%). (i) HCTNet (59.53%). (j) ExViT (72.98%). (k) FDNet (76.65%).

Augsburg, and MUUFL datasets, respectively. Such remarkable advancements highlight the effectiveness of our method in precisely identifying various land cover characteristics within complex scenes.

For the Houston2013 dataset, which includes 15 different land cover categories and has high scene complexity due to some areas being obscured by clouds, we can draw the following conclusions from Table II. First, feature extraction before multimodal data fusion is crucial for using feature-level fusion methods. EndNet, based on deep encoder–decoder network architecture, forces the fusion of features to sequentially reconstruct multimodal inputs and fuse multimodal information. This fusion strategy activates cross-modal neurons in a simple yet effective way, allowing for a more efficient fusion of HSI and LiDAR data, achieving higher classification scores than CoupledCNN, with an improvement of 6.28% in OA. In comparison to EndNet, MS2CANet establishes the spatial correspondence between HSI and LiDAR features through the spatial–spectral cross-modal attention (S2CA) module, extracts multiscale features via a multiscale pyramid convolution, and reduces irrelevant information on each scale using an effective feature recalibration (EFR) module. This strategy of enhancing multimodal features with spatial information is superior to direct addition or weighted fusion, achieving an improvement of approximately 1.1% in OA compared to EndNet. Different from methods that focus on improving feature fusion, S2ENet explores how to effectively promote the interaction of complementary information in the HSI-LiDAR data before fusion. The proposed spatial enhancement module (SAEM) enhances the spatial representation of HSI features and the spectral representation of LiDAR features. Therefore, S2ENet achieves an improvement of 6.34% in OA compared to MS2CANet and achieves 100% classification accuracy in four land cover categories, validating the efficacy of the method in seamlessly integrating the spatial–spectral information from hyperspectral data with the elevation information obtained from LiDAR. Second, compared to methods based on deep CNN architectures, CNN-transformer-based models

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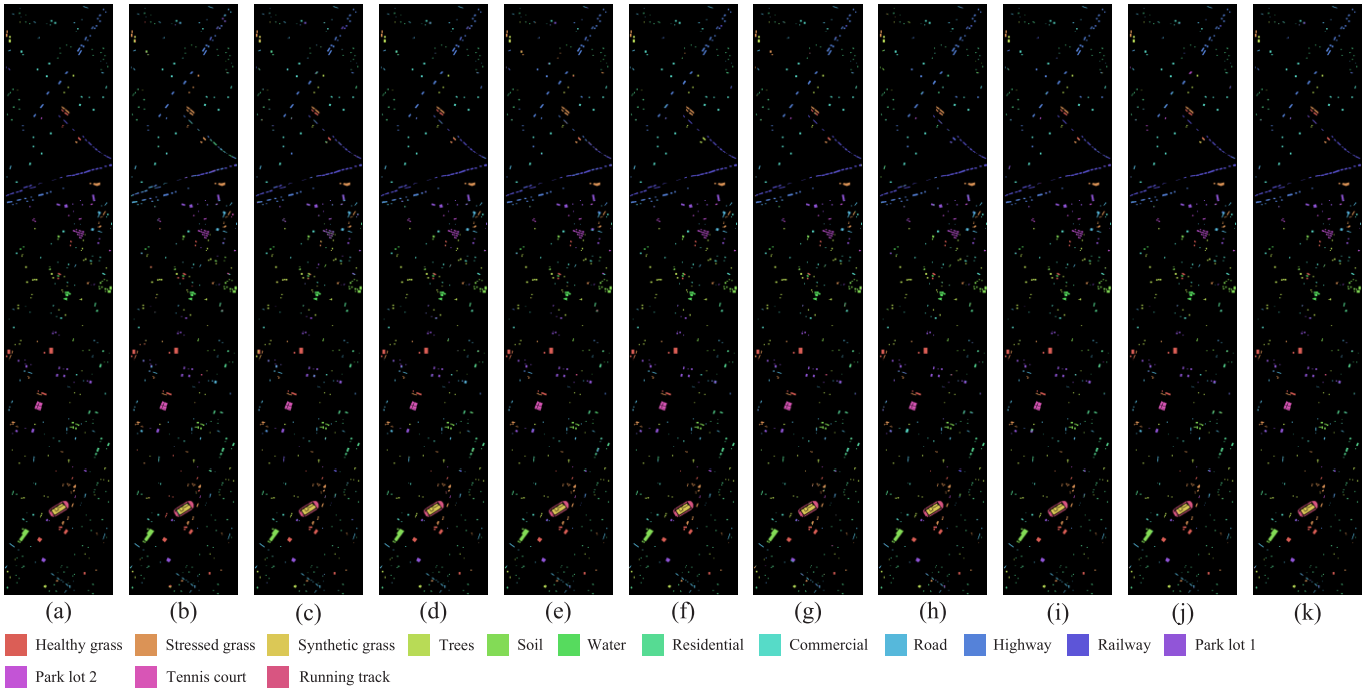


Fig. 9. Classification maps by different methods on the Houston2013 dataset. (a) Ground-truth. (b) CoupledCNN (81.30%). (c) EndNet (87.58%). (d) MDL-Cross (92.31%). (e) MS2CANet (88.64%). (f) S2ENet (94.98%). (g) MFT (93.11%). (h) NNCNet (90.61%). (i) HCTNet (94.53%). (j) ExViT (94.65%). (k) FDNet (95.91%).

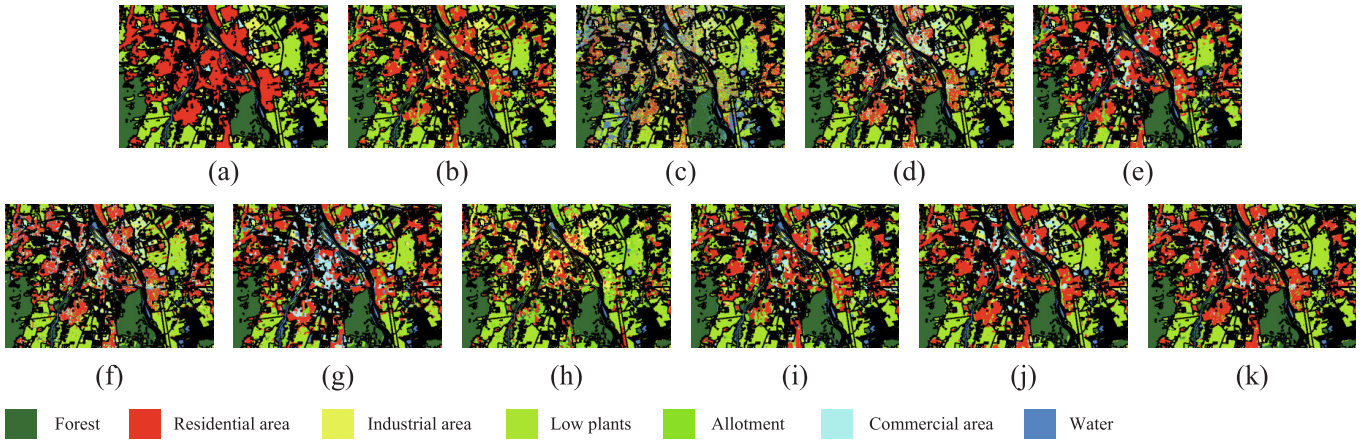


Fig. 10. Classification maps by different methods on the Augsburg dataset. (a) Ground-truth. (b) CoupledCNN (75.83%). (c) EndNet (56.58%). (d) MDL-Cross (71.84%). (e) MS2CANet (69.29%). (f) S2ENet (75.61%). (g) MFT (67.99%). (h) NNCNet (57.47%). (i) HCTNet (76.99%). (j) ExViT (79.35%). (k) FDNet (82.30%).

can effectively utilize global features, thereby improving the model's classification performance. The multihead cross patch attention (mCrossPA) in MFT, including global class tokens and HSI patch tokens, enhances the model's ability to learn unique representations in the hierarchical feature space. Therefore, compared to the well-performing MDL-Cross based on deep CNN architectures, MFT shows better classification performance, with an improvement of 0.8% in OA. In addition, ExViT enhances the exchange of multimodal information through a cross-modality attention (CMA) module, improving the discriminative ability of classification tokens in each modality, achieving a classification accuracy of approximately 2.2% higher than MDL-Cross. Third, 3-D CNNs have a better ability to learn spectral information in HSI data compared to 2-D CNNs. The deep hierarchical architecture designed by HCTNet fully utilizes the powerful spatial contextual extraction ability of CNNs and leverages the characteristic of 3-D CNNs to simultaneously capture spatial and spectral features, locally modeling the high-dimensional spectral information to capture spatial–spectral features from HSI data. Therefore, compared to ExViT, which has a similar CNNs-transformer structure and only uses 2-D CNNs, HCTNet achieves higher per-class accuracy of around 2% and 2.5% on two land cover categories with similar materials, "Park lot 1" and "Park lot 2," respectively. This is due to the deeper exploration of spatial–spectral features in HSI data by

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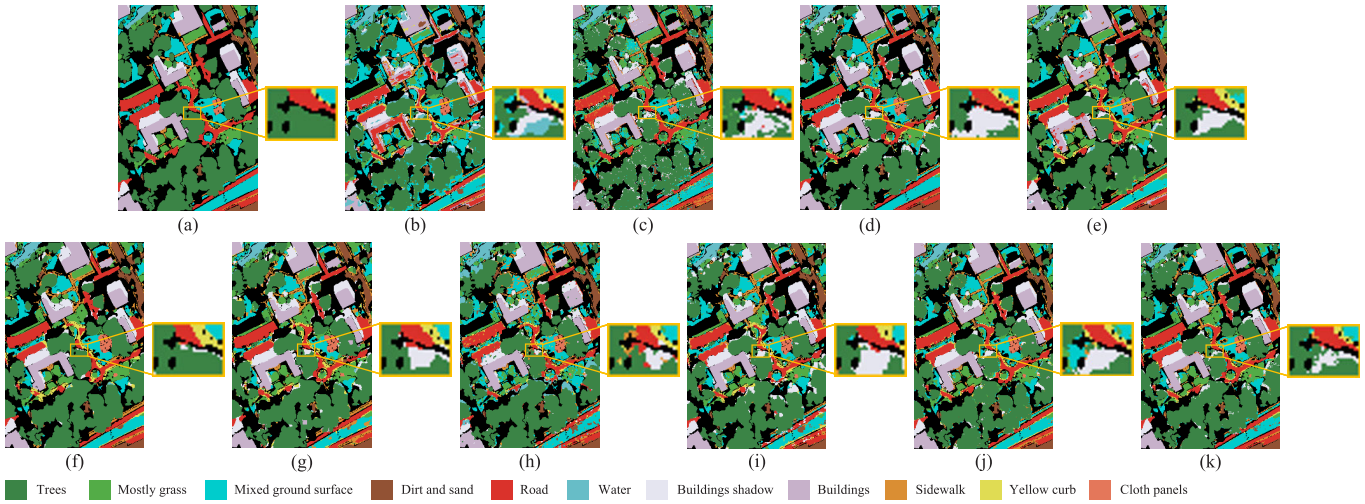


Fig. 11. Classification maps by different methods on MUUFL dataset. (a) Ground-truth. (b) CoupledCNN (73.42%). (c) EndNet (81.46%). (d) MDL-Cross (84.77%). (e) MS2CANet (83.53%). (f) S2ENet (86.25%). (g) MFT (83.08%). (h) NNCNet (81.06%). (i) HCTNet (82.94%). (j) ExViT (85.13%). (k) FDNet (88.12%).

3-D CNNs, enabling the extraction of distinguishable detailed features and achieving more accurate classification. Fourth, preprocessing the input HSI and LiDAR data is beneficial for the model to learn robust deep features. The proposed FDNet adopts a DWTS module to separate the high and low-frequency components of the input multimodal data in the frequency domain, contributing to the global processing of the overall information of the low-frequency components and the enhancement of detailed information in the high-frequency components. Compared to HCTNet with a dual-branch structure based on hierarchical CNNs and transformer, FDNet, which has a similar structure and independently processes high and low-frequency components, achieved better classification performance, with an improvement of about 1.4% in OA. Therefore, a simple and efficient CNN-based network is utilized to learn the spatial–spectral features of the HSI data and the elevation features of the LiDAR data from the input high and low-frequency components before the feature fusion stage. Then, through a feature-level fusion strategy based on weighted operations, the learned features are effectively fused, followed by the designed FAE module for rapid global feature learning, thereby boosting the model's classification performance. Experimental results show that the proposed FDNet obtains the best classification performance for OA, AA, and Kappa on the Houston2013 dataset, and achieves the highest classification scores in 8 out of 15 land cover categories.

Table I lists the quantitative classification results of Yancheng data. FDNet obtains the highest classification accuracy on all three evaluation metrics, achieving the highest accuracy in 3 out of 9 land covers. Table III illustrates the quantitative classification results of Augsburg data, and the proposed method acquires the highest classification level in 1 out of 7 categories. Table IV displays the quantitative classification results of MUUFL data, with our model achieving the best performance in terms of OA, AA, and Kappa, surpassing the second-best model by approximately 1.9%, 1.2%, and 2.4%, respectively, and achieving the highest per-class accuracy in 4 out of 11 categories.

2) Visual Comparison and Analysis: Figs. 8–11 illustrates the visual classification results of several comparison methods on the Yancheng, Houston2013, Augsburg, and MUUFL datasets. The MUUFL dataset utilizes local magnification to clearly show the differences in classification performance between different models. It can be seen that the visual classification results of the proposed FDNet are closer to the ground-truth map, producing smoother classification results and demonstrating better classification performance. On the other hand, methods, i.e., EndNet and MDL-Cross, have more isolated points. As shown in Fig. 8, most methods incorrectly classify some "S. alterniflora" as other categories like "Grass" in the Yancheng dataset. Especially, the proposed FDNet can effectively capture deeper features from the HSI-LiDAR data, more accurately depicting various categories in the scene, resulting in fewer noisy visual classification results. As illustrated in the zoomed-in view area of Fig. 11, the boundaries between the "Trees" and "Road" classes produced by other methods are not only difficult to distinguish but also lead to more classification errors. For example, CoupledCNN and ExViT incorrectly classify some "Trees" as "Water," while MDL-Cross and MS2CANet incorrectly classify numbers of "Trees" as "Buildings shadow." It is worth noting that the proposed method's classification results are closer to the ground-truth map, generating clearer classification boundaries and fewer misclassified samples. Toward the end, FDNet demonstrates strong classification performance, fully proving its advancement and effectiveness through quantitative and visual analysis.

3) Intermediate Visual Comparison: To provide a more intuitive visual comparison, we adopt the t-SNE method to visualize the feature distribution of the proposed model and various advanced methods on the Houston2013 dataset, as shown in Fig. 12. Notably, the visualization of feature

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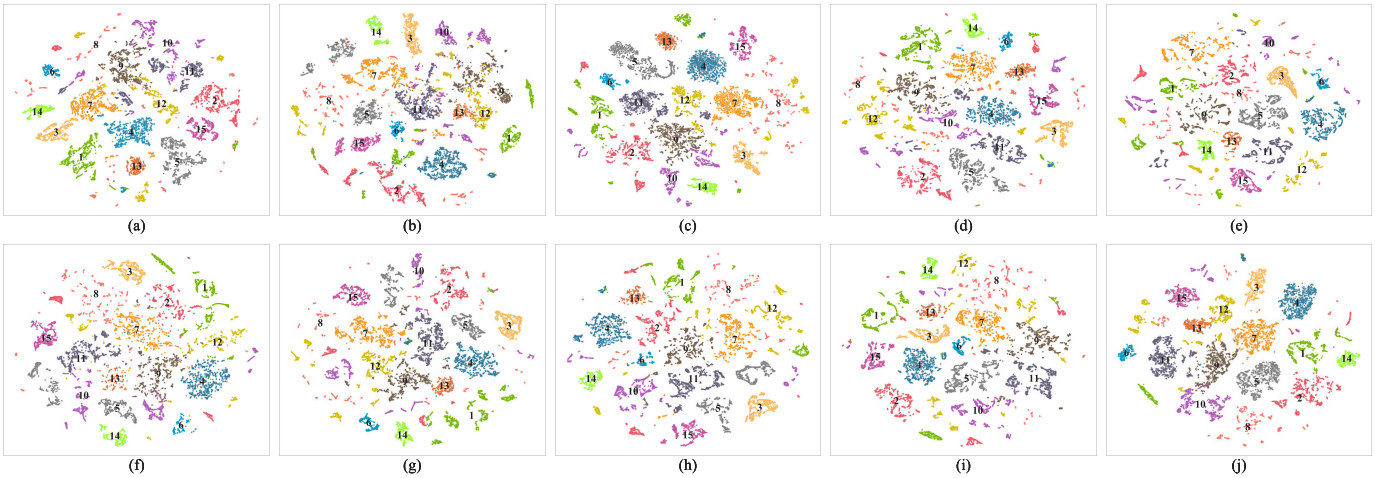


Fig. 12. t-SNE visualization of feature distribution in different methods on Houston2013 dataset. (a) CoupledCNN. (b) EndNet. (c) MDL-Cross. (d) MS2CANet. (e) S2ENet. (f) MFT. (g) NNCNet. (h) HCTNet. (i) ExViT. (j) FDNet.

TABLE V
COMPARISON OF MODEL COMPLEXITY OF DIFFERENT METHODS ON HOUSTON2013 DATASET

Methods	CoupledCNN	EndNet	MDL-Cross	MS2CANet	S2ENet	MFT	NNCNet	HCTNet	ExViT	FDNet(ViT)	FDNet(FAE)
Trainable Parameters	104.78K	91.97K	119.10K	179.25K	270.80K	312.43K	3.87M	429.80K	229.10K	544.77K	345.86K
Computation Cost(MFLOPs)	2.48	0.46	6.72	2.23	65.43	50.32	77.81	7.93	3510.00	14.86	11.68

distribution almost matches the quantitative results. For instance, for the "No.8 Commercial" category with lower per-class accuracy in most methods, the distribution of its corresponding features is more scattered compared to other categories; whereas for the "No.14 Tennis court" category, which is almost perfectly classified in all methods, the distribution of corresponding features is more compact with hardly any isolated feature points. In addition, the proposed FDNet overall has fewer isolated feature points, and the feature distribution of the same category is more concentrated and compact. On the contrary, other classification methods with lower accuracy, such as CoupledCNN and EndNet, have more scattered feature distributions, and the boundaries between different category features are more blurred. Therefore, intermediate visual comparison results demonstrate the superiority of FDNet.

4) Computational Complexity Analysis: Table V presents the computational complexity of different methods on the Houston2013 dataset, as reflected by the trainable parameters during the back-propagation stage and million floating-point operations (MFLOPs). It can be observed that EndNet, as a lightweight network, has the fewest trainable parameters and MFLOPs, while NNCNet with a bilinear AM utilizes the most parameters. Due to the adoption of spatial–spectral enhancement module, S2E acquires the highest number of parameters among all CNNs-based models and surpasses four CNNs-Transformer-based models in MFLOPs. Furthermore, CNNs-Transformer-based models generally have more trainable parameters compared to CNNs-based models, mainly due to the extensive parameters in the attention mechanism of the transformer. However, the proposed FDNet (FAE) exhibits relatively lower computational complexity as compared to NNCNet and HCTNet with similar CNNs-transformer architecture.

In addition, to further demonstrate the superiority of the proposed FDNet in computational efficiency, while ensuring a certain level of classification performance, the FAE module is replaced with a conventional ViT of the same depth. Based on experimental results, it can be seen that the variant FDNet (ViT) contains more MFLOPs and trainable parameters compared to the proposed FDNet (FAE), demonstrating the effectiveness of the designed FAE in reducing computational complexity. Therefore, FDNet exhibits relatively high computational efficiency and optimal classification performance in an acceptable computational costs.

D. Parameters Analysis

1) Type of the Wavelet: For the DWT used in our network, the wavelet basis functions, including their type and order, may have an impact on classification results. We analyze the performance of Haar, Daubechies, and Biorthogonal wavelets on four datasets as shown in Table VI, where "db$x$" and "bior$x.y$" represent Daubechies wavelet with order x and Biorthogonal wavelet with orders (x and y), respectively, the best classification results are highlighted in bold. The length of the wavelet filters increases with the order and encompasses more filter coefficients. From the table, it can be observed that the best classification performance is generally achieved using Daubechies and Biorthogonal wavelets compared to the Haar wavelet. This could be attributed to the fact that the

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TABLE VI
EFFECT OF WAVELET BASIS FUNCTIONS FOR DWT ON CLASSIFICATION PERFORMANCE (BEST RESULTS ARE IN BOLD)

Wavelet		Yancheng			Houston2013			Augsburg			MUUFL		
		OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa
Haar		71.30	78.31	64.31	94.47	95.30	94.02	85.54	75.77	80.15	87.09	87.92	83.25
Daubechies	db2	76.65	83.15	71.03	95.91	96.13	95.58	82.30	74.74	76.00	**88.12**	**89.31**	**84.58**
	db3	70.18	77.03	62.81	94.35	94.97	93.89	80.11	72.99	73.33	87.26	87.84	83.47
	db4	75.71	81.53	69.86	93.96	94.63	93.47	77.08	70.02	69.64	86.34	86.17	82.21
Biorthogonal	bior1.1	**77.72**	**83.57**	**72.21**	**96.23**	**96.59**	**95.93**	**86.98**	**77.79**	**82.06**	85.44	87.02	81.22
	bior2.2	76.30	83.09	70.59	93.72	94.53	93.21	42.87	58.10	33.69	87.77	89.31	84.16
	bior4.4	75.26	81.79	69.22	94.05	94.76	93.57	82.44	75.15	76.23	87.73	89.47	84.10

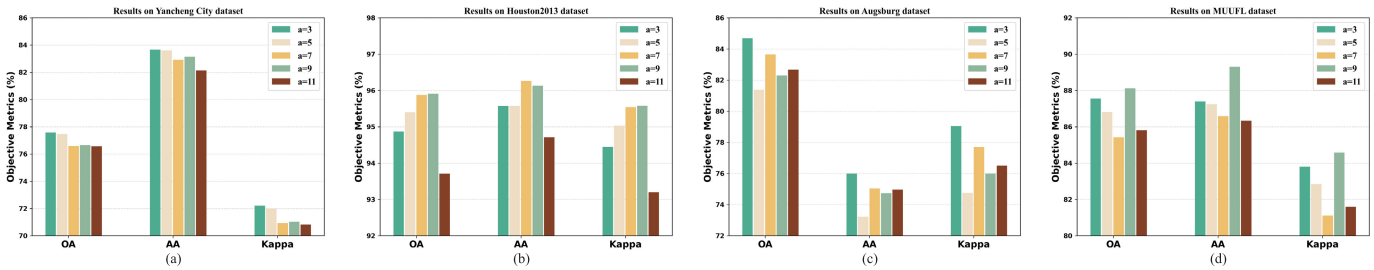


Fig. 13. Effect of kernel size a in SA on classification performance. (a) Yancheng. (b) Houston2013. (c) Augsburg. (d) MUUFL.

Haar wavelet is the simplest wavelet basis function, while Daubechies and Biorthogonal wavelets offer more flexibility and better frequency characteristics, allowing for a more precise capture of detail features. In addition, Biorthogonal wavelets outperform Daubechies wavelets in the Yancheng, Houston2013, and Augsburg datasets, achieving approximately 1%, 0.3%, and 4.7% higher classification accuracy in terms of OA, respectively. This could be attributed to the unique symmetry and orthogonality of Biorthogonal wavelets, ensuring minimal redundancy between the basis functions of the wavelet transform and thus more effectively representing the information in HSI-LiDAR data. Despite higher-order wavelet basis functions having better localization properties, the results indicate that wavelets with shorter filters ("db2" and "bior1.1") achieve higher classification accuracies relative to those with longer filters ("db4" and "bior4.4"). Therefore, the "db2" wavelet is adopted as the basis function for DWT with the consideration of both the increase in computational complexity with the length of the wavelet basis functions and the consistency of the model structure.

2) Kernel Size of SA: We discuss the impact of convolutional kernel size a in spatial AM. Specifically, we adjust the kernel size to 3×3, 5×5, 7×7, 9×9, and 11×11, respectively. The classification performance of FDNet is evaluated on four datasets without changing other parameters. The classification results are shown in Fig. 13. It can be observed that different kernel sizes exhibit certain differences in classification performance across the four datasets, which may be attributed to variations in scene complexity among the datasets. For datasets with relatively more land cover classes such as Houston2013 and MUUFL, larger kernel size ($a = 9$) achieves second-best and best classification accuracy, respectively, while excessively large kernel size ($a = 11$) may lead to a decline in classification performance due to the model learning redundant features. In addition, for more complex scenes, small kernel sizes limit the model's ability to extract sufficient spatial contextual information, resulting in poorer classification performance. For datasets with relatively fewer land cover classes such as Yancheng and Augsburg, smaller kernel sizes ($a = 3$) demonstrate the best classification performance. Based on the results from the four datasets, we set the kernel size to 9 to ensure the uniformity of the model structure.

3) Weight Coefficient for Feature Fusion: In this section, we explore the impact of fusion coefficient ω between HSI features and LiDAR features in the S2FFE module. While keeping other parameters constant, we set the value of ω from 0.1 to 0.9 in fixed increments of 0.2 on four datasets to acquire the network's classification performance. As depicted in Fig. 14, while ω is too small ($\omega = 0.1$), the model's classification performance is not satisfactory, achieving the lowest accuracy on two datasets, which can be attributed to the lack of spectral information from HSI in the fused features. Furthermore, as ω is relatively large ($\omega = 0.9$), the proposed model yields lower classification results on all datasets, with the lowest scores on the MUUFL dataset, possibly due to the model potentially focusing too much on the spectral features of HSI data while neglecting to learn the elevation features present in LiDAR data. Moreover, while ω is at an intermediate value ($\omega = 0.5$), the model achieves

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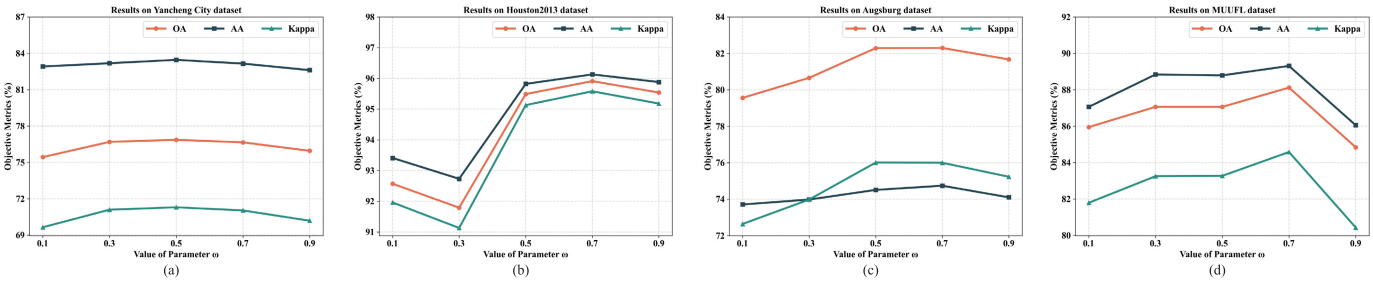


Fig. 14. Effect of the feature fusion weight coefficient ω on classification performance. (a) Yancheng. (b) Houston2013. (c) Augsburg. (d) MUUFL.

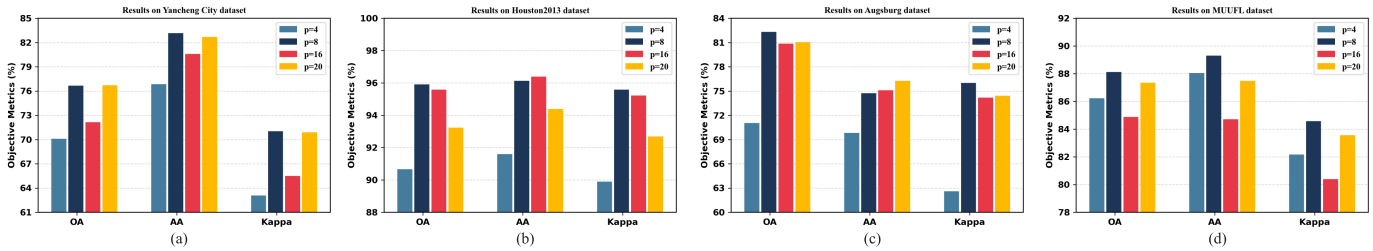


Fig. 15. Effect of the input patch size p on classification performance. (a) Yancheng. (b) Houston2013. (c) Augsburg. (d) MUUFL.

TABLE VII
ABLATION RESULTS ON AM AND FAE

		Yancheng			Houston2013			Augsburg			MUUFL		
AM	FAE	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa
✗	✗	73.77	81.55	67.74	91.71	92.61	91.03	79.69	73.11	72.60	84.44	82.16	79.59
✗	✓	73.79	82.03	67.83	95.56	96.06	95.20	80.03	74.25	73.23	87.17	88.70	83.37
✓	✗	76.39	83.05	70.72	95.29	95.61	94.90	81.79	74.68	75.39	87.47	89.23	83.78
✓	✓	**76.65**	**83.15**	**71.03**	**95.91**	**96.13**	**95.58**	**82.30**	**74.74**	**76.00**	**88.12**	**89.31**	**84.58**

the best or second-best performance on all datasets, reflecting the importance of utilizing features from multimodal data. However, for Houston2013 and MUUFL with more land cover categories, the fusion weight of 0.7 has a better classification performance, the main reason is that HSI data contains richer features to distinguish categories in more complex scenes. Therefore, the fusion coefficient value of 0.7 is selected based on the classification results of four datasets.

4) Input Patch Size: As shown in Fig. 15, we analyze the impact of different input patch sizes on classification performance. Specifically, we adjust the input patch size to 4, 8, 16, and 20 while keeping other parameters fixed, and represent the corresponding classification accuracies with blue, dark blue, red, and yellow bars, respectively. It can be concluded that both too-small ($p = 4$) and too-large ($p = 20$) input patch sizes lead to a decrease in performance for four datasets. This may be because too small patch sizes lack sufficient input information, resulting in a lack of necessary contextual information, thus failing to fully capture important features in HSI and LiDAR data. Conversely, too large patch sizes may lead to learning redundant features, while the model may fail to effectively capture local detailed features, thereby affecting classification accuracy. However, an input patch size ($p = 8$) achieves the highest scores on all four datasets in OA, AA, and Kappa. Then, the input patch size is set to 8 to acquire the high classification scores.

E. Ablation Studies

In FDNet, AM in S2FFE and FAE are employed in the low-frequency feature and global feature extraction stages, respectively. Herein, we conduct the ablation experiment to specifically validate the effectiveness of AM and FAE. Based on three objective evaluation metrics: OA, AA, and Kappa, Table VII presents the experimental results from four datasets. From the results, one can observe that the proposed FDNet achieves the highest classification accuracy on all datasets. For the Yancheng dataset, the absence of AM and FAE leads to reductions of 2.86% and 0.26% in OA, respectively. The variants without AM and without FAE experience decreases of 0.35% and 0.62% in OA for the Houston2013 dataset, respectively. For the Augsburg dataset, the absence of AM and FAE lead to reductions of 2.27% and 0.51% in OA, respectively. As for the MUUFL dataset, the absence of AM and FAE brings about decreases in 0.95% and 0.65% in OA, respectively. These results demonstrate the effectiveness of AM in adaptively learning important features and the robust learning capability of the designed FAE for global spectral features. Furthermore, while AM and FAE are both not adopted,

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the network achieves the lowest classification scores on all four datasets, which also confirms the effectiveness of AM and FAE modules independently in improving the model's classification accuracy.

V. Conclusion

This article proposes a global–local feature deep learning model, called FDNet, for joint classification of HSI and LiDAR data based on frequency feature learning style. A three-branch module based on the DWT method, named S2FFE, is utilized to learn overall spatial features in the low-frequency component, adaptively extract detailed information in the high-frequency component, and acquire HSI-LiDAR fusion features in a weighted manner. Subsequently, the designed FAE, including FFT and convolutional offsets, captures global features in the frequency domain and effectively enhances computational efficiency. Finally, a CNN-based network leverages the fusion features for classification, yielding the final classification results. Extensive experiments are conducted on four datasets, including the challenge Yancheng dataset. Quantitative and visual analysis suggests the superiority of the proposed FDNet over other related networks. Furthermore, the fusion strategy for multimodal data is crucial for the classification of HSI and LiDAR data. Therefore, the main research content in the future is to develop a more effective adaptive multimodal feature fusion mechanism. In order to better integrate low and high-frequency features, another future research content of ours is to develop a fusion method that can dynamically adjust the importance of low and high-frequency features based on data characteristics, such as weighted averaging or allocating different attention weights in channel dimension.

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